Government Engineering College, Sheikhpura

Mid Term Examination- Dec 2025

100111- Programming for Problem Solving **(CSE Sec A and CSE Sec B)**

Full Marks-20 Time: 2 hrs

Section-A

Answer any five Questions **1 x 5 = 5**

1. What is the output of the following code :

```
int a = 5, b=3;
printf("%d %d", a, b);
```

(a) 5 2 (b) 7 2

(c) 6 2 (d) 6 3

2. Which logical operator returns true only if both operands return true :

(a) `||` (b) `&&`

(c) `|` (d) `^`

3. Which of the following is valid variable name in C ?

(a) `int` (b) `_temp123`

(c) `123var` (d) `var-name`

4. What is the correct format specifier to print a float number?

(a) % number (b) % d

(c) % f (d) % (float number)

5. What is the output of following loop ?

```
for(int i = 0; i < 5; i++) {
    if(i == 3)
        break;
    printf("%d ", i);
}
```

(a) 0 1 2 (b) 0 1 2 3

(c) 0 1 2 3 4 (d) 0 1 2 4

6. Which of the following is **incorrect** way to initialize an integer array of size 5 with all zeros.

(a) int arr[5] = {0}; (b) int arr[5] = {all zero};

(c) int arr[] = {0,0,0,0,0}; (d) int arr[5] = {00000};

7. What is the role of **`continue`** statement in a loop ?

a) Exits the loop completely

b) Skips the current iteration and proceeds to the next

c) Restarts the loop from the beginning

d) Pauses the loop temporarily

Section-B

Answer any five Questions **3 x 5 = 15**

1. Differentiate between the different types of loop constructs in the context of C programming with examples.
2. Explain Relational and Bitwise operators with example programs
3. Define implicit and explicit type casting in C. Provide code examples illustrating when each type of casting occurs and their potential side effects.
4. Differentiate between conditional operator and conditional statements in C.
5. Write a C program to swap two integers :
   a. Using temporary variable.
   b. Without using temporary variable.
6. Write a C program to check if a given year is a Leap year or NOT.
7. Write a C program where user will enter a character and the program will detect if the character is a vowel or NOT.

x